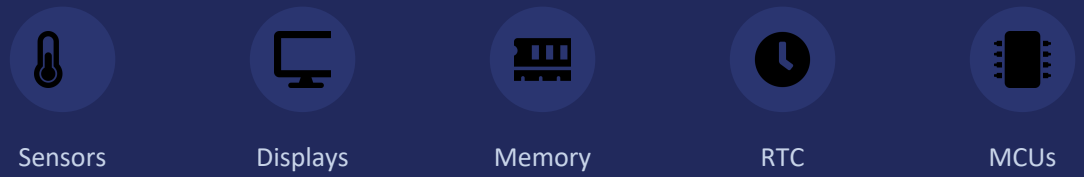
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EMBEDDED SYSTEMS • SERIAL COMMUNICATION

I²C Communication Protocol

I²C

Welcome to our presentation



Introduction to Communication Protocol

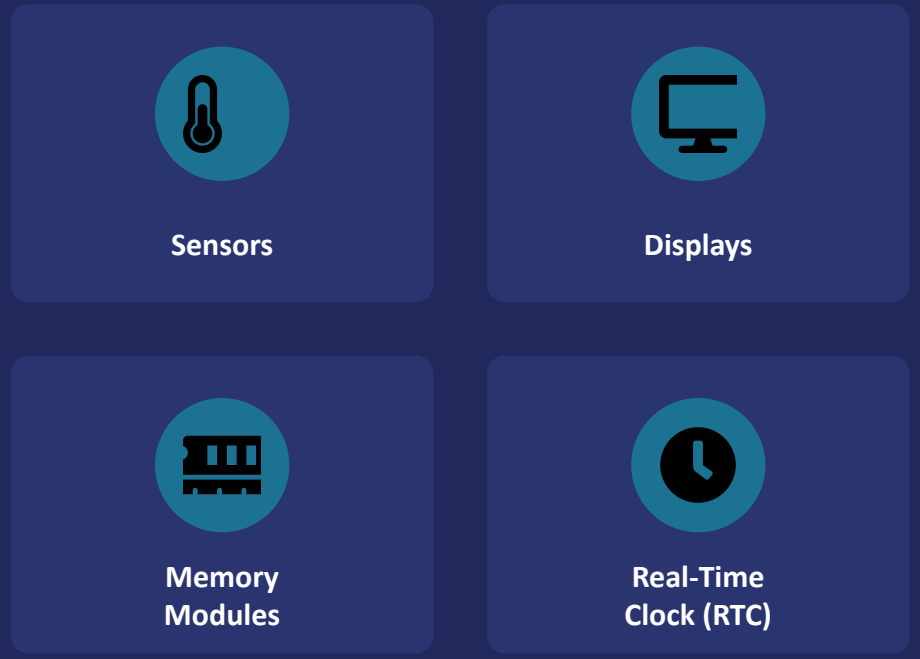
A microcontroller cannot work alone. It needs to communicate with external devices such as sensors, displays, memory modules, and real-time clocks.

To enable this communication, different communication protocols are used:

UART SPI I²C

COMMUNICATION PROTOCOL

A set of rules that allows two or more electronic devices to exchange data reliably.



MCU ↔ Peripheral Communication

Why I²C?

Comparing communication protocols for multi-device systems

Limitations of USART

- Supports only point-to-point communication
- Not suitable for connecting multiple peripheral devices

Limitations of SPI

- Requires multiple communication lines
- Each slave requires a dedicated Chip Select (CS) pin
- Increased wiring complexity for multiple devices

Why I²C?

- Supports multiple devices
- Uses only two wires
- No dedicated Chip Select (CS) line required
- Simple and efficient hardware design

What is I²C?

I²C (Inter-Integrated Circuit) is a synchronous serial communication protocol used for communication between multiple ICs using only two wires.

KEY TERMS

Serial Communication

Data is transmitted bit by bit between devices.

Synchronous Communication

Data transfer is synchronized using a shared clock signal.

Two-Wire Communication

Uses only two lines to connect all devices on the bus:

SDA

Serial Data Line

SCL

Serial Clock Line

SDA carries data; SCL provides the shared timing signal that keeps every device on the bus in sync.

Where is I²C Used?

I²C is widely used for communication between microcontrollers and various peripheral devices.



Display

- OLED Displays



Sensors

- Temperature Sensors
- Pressure Sensors
- Accelerometer & Gyroscope Modules



Memory

- EEPROM

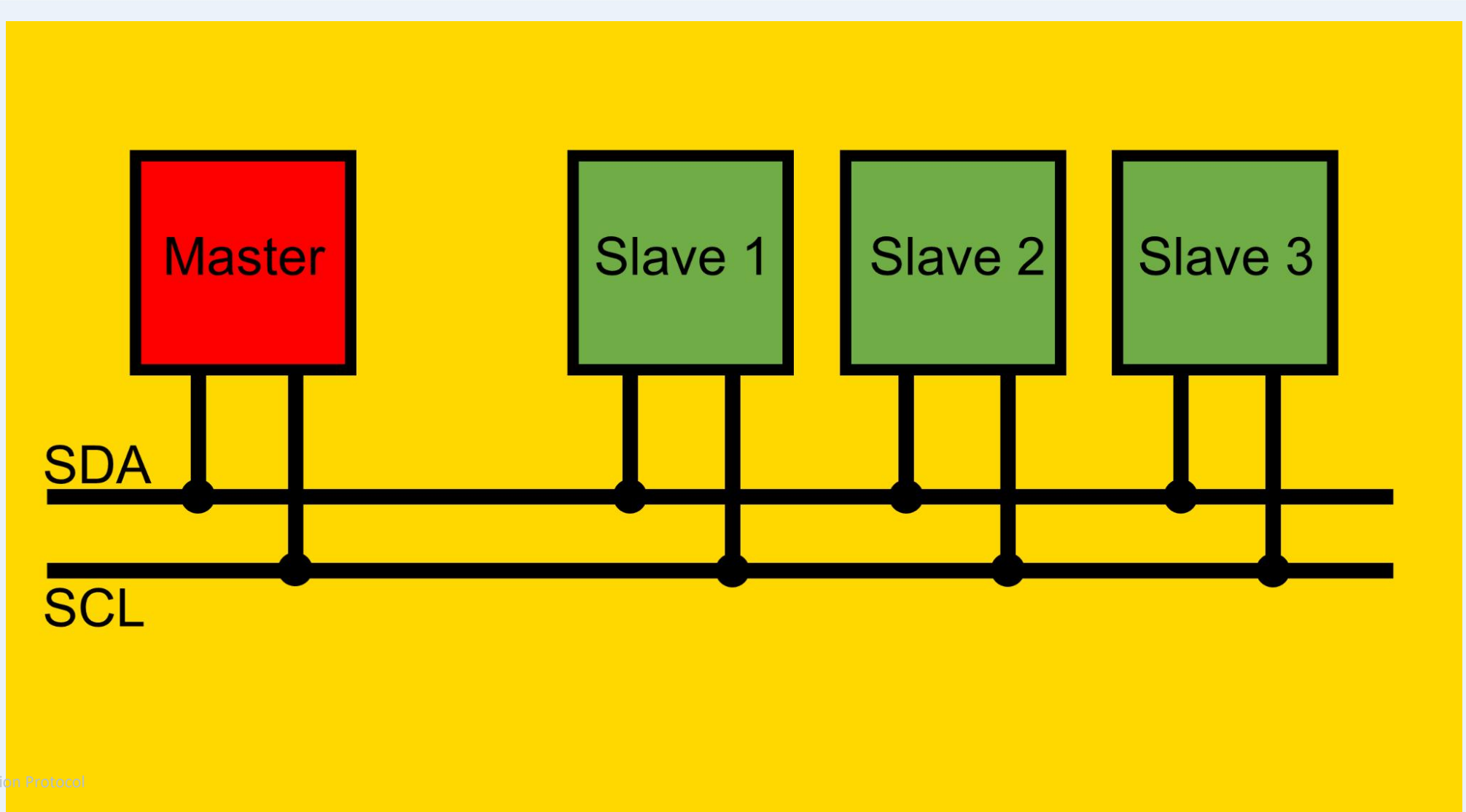


Time Keeping

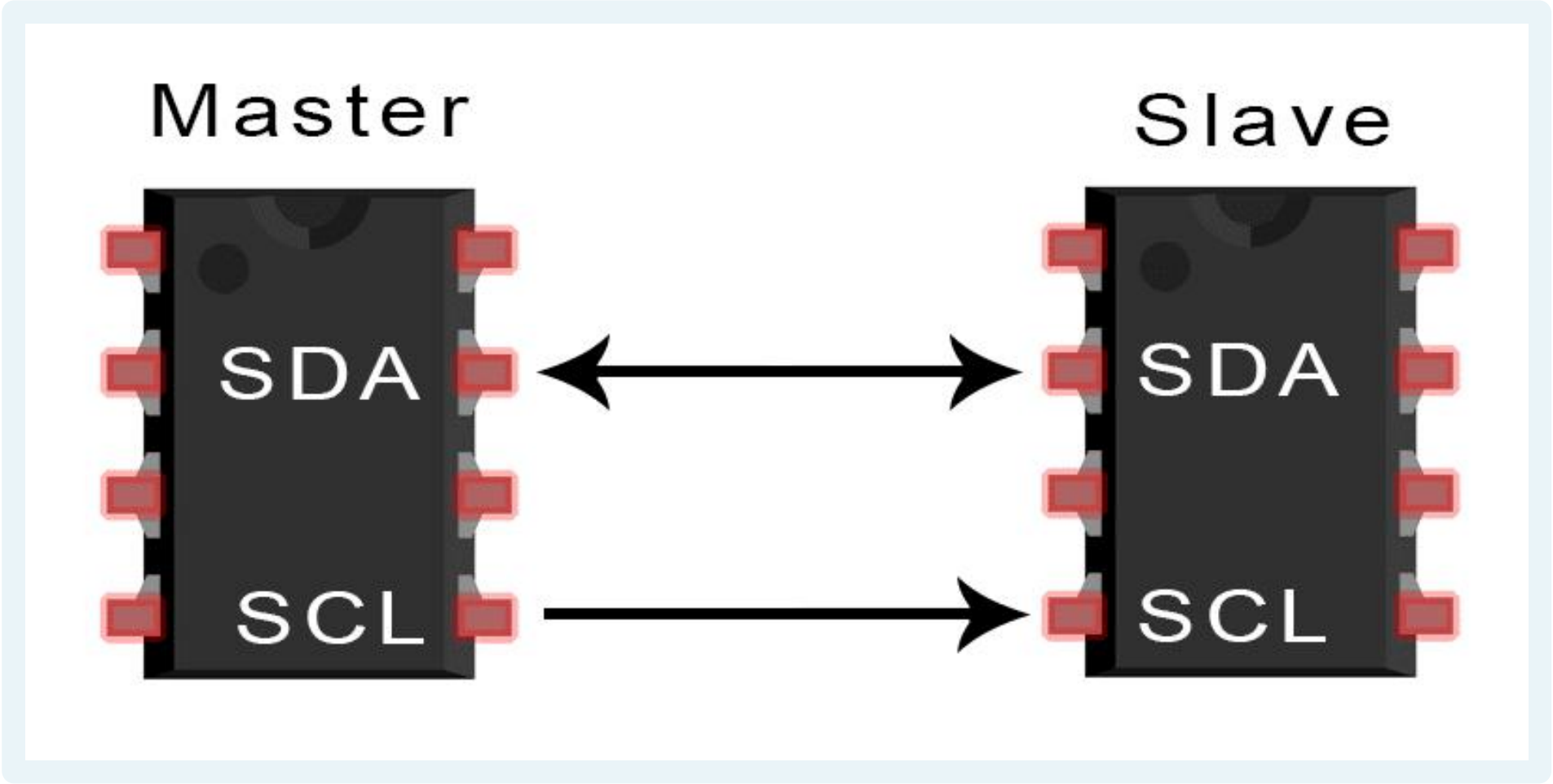
- Real-Time Clock (RTC)

Basic Architecture of I²C

How a master microcontroller connects to multiple slave devices over the I²C bus



I²C Bus Concept



SDA

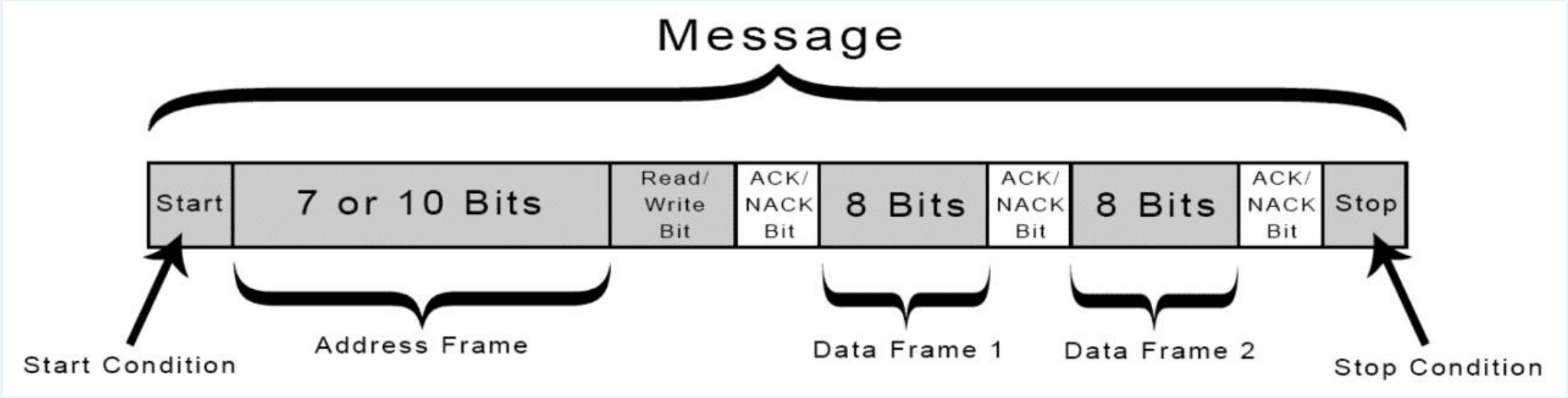
SDA (Serial Data) – The line for the master and slave to send and receive data.

SCL

SCL (Serial Clock) – The line that carries the clock signal.

How I²C Works

I²C transfers data in the form of messages, divided into smaller units called frames.



A MESSAGE CONTAINS

Address Frame → Contains the address of the target slave device

Data Frame(s) → Contains the actual data being transmitted

CONTROL BITS MANAGE COMMUNICATION

Start Condition
Begins communication

Read/Write Bit
Defines data direction

ACK/NACK Bit
Confirms successful reception

Stop Condition
Ends communication

I²C Communication Control Signals

Two signal transitions frame every I²C transaction on the bus

Start Condition

- Indicates the beginning of communication
- SDA changes from HIGH → LOW while SCL remains HIGH

Stop Condition

- Indicates the end of communication
- SDA changes from LOW → HIGH after SCL goes HIGH

I²C Communication Control Bits

Three control fields govern addressing and acknowledgment on the bus



Address Frame

- Contains the 7-bit or 10-bit address of the target slave
- Helps the master select a specific device



Read/Write Bit

- Defines the direction of communication
- 0 → Write data to slave
- 1 → Read data from slave



ACK/NACK Bit

- Confirms whether data is received successfully
- ACK → Successful reception
- NACK → No acknowledgement

I²C Addressing Communication Process

How the master locates and confirms the correct slave device on the bus

1

STEP

Master sends the address of the target slave device

2

STEP

All connected slave devices receive and compare the address

3

STEP

The matching slave sends an ACK signal to the master

4

STEP

Other slaves ignore the message and remain inactive

The Data Frame

How actual data moves across the bus once addressing is complete

- Actual data transfer begins after receiving ACK
- Each data frame contains 8-bit data
- Data is transmitted MSB first
- Receiver sends ACK/NACK after each data frame
- ACK confirms successful data reception
- Master sends STOP condition after completing the transfer

Advantages & Disadvantages of I^2C

Advantages

- Requires only two communication wires (SDA & SCL)
- Supports multiple masters and multiple slave devices
- ACK/NACK mechanism ensures reliable data transfer
- Simplifies hardware design compared to UART-based systems
- Widely used and well-established communication protocol

Disadvantages

- Lower data transfer speed compared to SPI
- Data frame size is limited to 8 bits
- More complex hardware implementation than SPI



Thank You

Thank you all